

NAME _____

TEACHER _____

Sydney Technical High School**2 Unit Mathematics****Year 11****Assessment task 2****August 2009****General Instructions**

- Each question attempted is to be started on a NEW PAGE, clearly marked with the number of the question, your name and class on the top right hand side of the page
- Working time allowed - 70 minutes
- Questions are of UNEQUAL value
- Write using black or blue pen
- APPROVED CALCULATORS may be used
- All necessary working should be shown. Marks may be deducted if working is poorly set out or difficult to read

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	TOTAL
/ 8	/ 8	/ 7	/ 7	/ 7	/ 8	/ 7	/ 7	/ 59

Question 1 (8 marks)

- a) Factorise fully : $2x^3 - x^2 + 8x - 4$ (2)
- b) Solve $-1 < 2x + 3 \leq 5$ (2)
- c) Find the domain and range of $y = \frac{1}{(2x + 3)}$ (2)
- d) Show that the points A(3,2) , B(-2,1) and C(8,3) are collinear (2)

Question 2 (8 marks) (Start a new page)

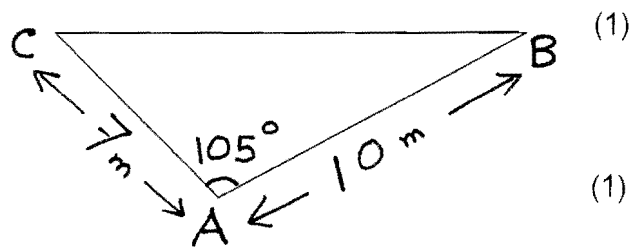
- a) Use the quadratic formula to solve the equation : $4x^2 + 5x - 2 = 0$
leaving your answers in the exact form (2)
- b) Simplify $5\sqrt{3} + \sqrt{20} - 2\sqrt{12} + \sqrt{45}$ (2)
- c) Simplify $\frac{\sin^2 \theta + \cos^2 \theta}{\tan^2 \theta}$ (1)
- d) Solve $\cos \theta = -\frac{1}{\sqrt{2}}$ for $0^\circ \leq \theta \leq 360^\circ$ (2)
- e) Is $f(x) = x^3 - x$ an odd function ? Explain your answer. (1)

Question 3 (7 marks) (Start a new page)

- a) The sum of the interior angles of a regular polygon is 2340° . Find the measure of each interior angle of the polygon. (1)
- b) Prove that $(1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$ (2)

- c) (i) Find the length of BC to the nearest cm.

- (ii) Find the Area of $\triangle ABC$
(in m^2 to 2 decimal places)



- d) Find the perpendicular distance from $(-2, 2)$ to the line $6x + 3y - 1 = 0$ in the exact form

Question 4 (7 Marks) (Start a new page)

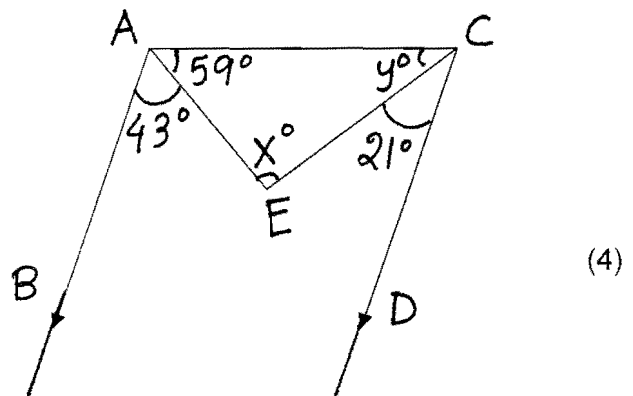
- a) Find 'x' if $\sin 80^\circ = \cos (90 - x)^\circ$ (1)

- b Draw a neat sketch of the following curve showing all relevant points

$$y = (x + 2)^2 + 2$$
 (2)

- c) Evaluate 'x' and 'y' giving reasons

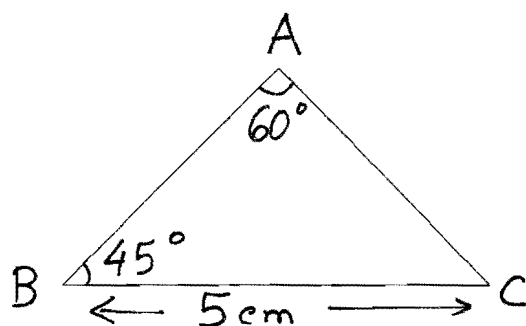
(drawing not to scale)



Question 5 (7 Marks) (Start a new page)

- a) Solve $|2x + 1| = 3x - 2$ and check solutions (3)

b)



Find the exact length of AC

(2)

c) Find the equation of the straight line that makes an angle of 135° with the positive x - axis and passes through the point (2,6)

(2)

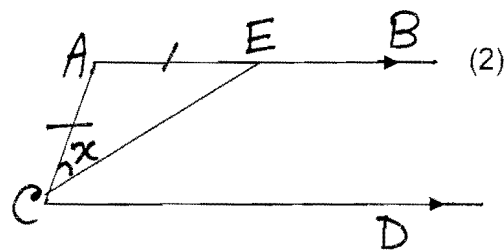
Question 6 (8 marks)

(Start a new page)

a) AB is parallel to CD and $AE = AC$

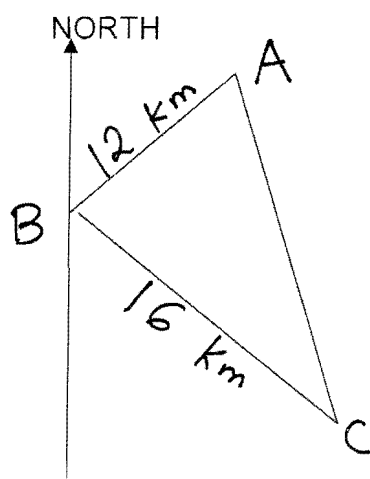
Let $\angle ACE = x$

Prove that $\angle ACE = \angle ECD$



(2)

b) Two yachts sail in a straight line from a buoy B. Yacht A sails 12 km in the direction 038° T and yacht C sails 16 km in the direction 128° T. Copy the diagram into your books and show all the angles given.



(i) How far apart are A and C ?

(2)

(ii) What is the bearing of yacht A as seen from yacht C, to the nearest minute?

(2)

(drawing not to scale)

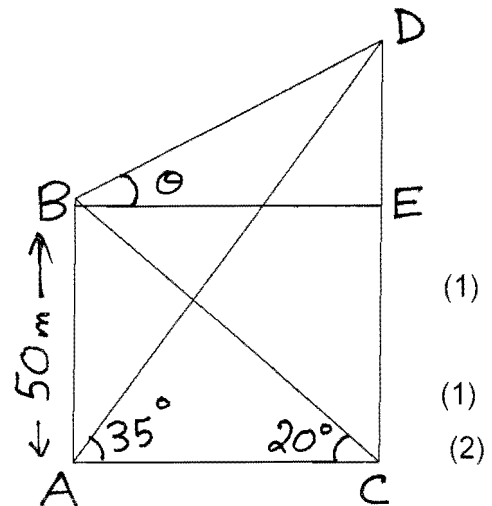
- c) Find the equation of the straight line passing through $(3, 7)$ and parallel to the line $5x - y - 2 = 0$ (2)

Question 7 (7 Marks)

(Start a new page)

- a) AB and CD are two vertical buildings with their bases A and C on level ground. The height of AB is 50m. The angle of elevation of B from C is 20° and angle of elevation of D from A is 35° . Calculate

- (i) Horizontal distance AC between the two buildings, to 1 decimal place
- (ii) The height of CD, to 1 decimal place
- (i) The angle of elevation θ of D as seen from B, to the nearest minute.



- b) Find the equation of the straight line with gradient of -2 that passes through the midpoint of the line joining $(5, -2)$ and $(-3, 4)$ (2)
- c) Find the exact value of $\cos 225^\circ$ (1)

Question 8 (7 marks)

(Start a new page)

- a) If $\sin x = -\frac{3}{5}$ and $\cos x > 0$, find the values of
 (i) $\tan x$ (ii) $\sec x$ (2)
- b) Solve : $\sqrt{3} \tan \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ (2)
- c) Prove that A $(1, 5)$, B $(4, -6)$ and C $(-3, -2)$ are the vertices of a right angled triangle (3)

TASK 2

2 UNIT MATHS YEAR 11

AUGUST 2009

Question 1

$$\begin{aligned} \text{a) } 2x^3 - x^2 + 8x - 4 \\ x^2(2x-1) + 4(2x-1) \\ = (2x-1)(x^2+4) \end{aligned}$$

$$\begin{aligned} \text{b) } -1 < 2x+3 \quad \& \quad 2x+3 \leq 5 \\ -4 < 2x \quad \quad \quad 2x \leq 2 \\ -2 < x \quad \quad \quad x \leq 1 \end{aligned}$$

$$\therefore -2 < x \leq 1$$

$$\begin{aligned} \text{c) If } 2x+3 = 0 \quad -\frac{3}{2} \\ 2x = -3 \\ x = -3/2 \end{aligned}$$

$\therefore$ Domain: all real x , $x \neq -3/2$
Range: all real y , $y \neq 0$

$$\text{d) } m_{AB} = \frac{2-1}{3-2} = \frac{1}{5}$$

$$m_{BC} = \frac{1-3}{-2-8} = \frac{-2}{-10} = \frac{1}{5}$$

since gradient equal A, B, C collinear

Question 2

$$\begin{aligned} \text{a) } x &= \frac{-5 \pm \sqrt{25 - 4 \times 4 \times -2}}{8} \\ &= \frac{-5 \pm \sqrt{57}}{8} \end{aligned}$$

$$\begin{aligned} \text{b) } 5\sqrt{3} + \sqrt{20} - 2\sqrt{12} + \sqrt{45} \\ = 5\sqrt{3} + 2\sqrt{5} - 4\sqrt{3} + 3\sqrt{5} \\ = \sqrt{3} + 5\sqrt{5} \end{aligned}$$

$$\begin{aligned} \text{c) } \frac{\sin^2 \theta + \cos^2 \theta}{\tan^2 \theta} &= \frac{1}{\tan^2 \theta} \\ &= \cot^2 \theta \end{aligned}$$

$$\begin{aligned} \text{d) } \cos \theta &= -\frac{1}{\sqrt{2}} \quad \begin{array}{c} \sqrt{s} | A \\ \sqrt{T} | C \end{array} \\ \text{acute } \theta &= 45^\circ \\ \therefore \theta &= 135^\circ, 225^\circ \end{aligned}$$

$$\text{e) } f(x) = x^3 - x$$

$$f(-x) = -x^3 + x$$

$$-f(-x) = x^3 - x$$

$$\therefore f(x) = -f(-x) \therefore \text{ odd fn}$$

Question 3

$$\text{a) int angle sum } 2340^\circ$$

$$(n-2) \times 180 = 2340$$

$$180n - 360 = 2340$$

$$180n = 2700$$

$$\text{no angles } n = 15$$

$$\therefore \text{ each interior angle } 156^\circ$$

$$\begin{aligned} \text{b) LHS} &= (1 - \tan^2 x)^2 + (1 + \tan^2 x)^2 \\ &= 1 - 2\tan^2 x + \tan^4 x + 1 + 2\tan^2 x + \tan^4 x \\ &= 2 + 2\tan^2 x \\ &= 2(1 + \tan^2 x) \\ &= 2\sec^2 x \\ &= \text{RHS} \end{aligned}$$

c) i)

$$BC^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \times \cos 105^\circ$$

$$BC = 13.61 \text{ m}$$

$$\therefore \text{ OR } BC = 1361 \text{ cm}$$

ii)

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 7 \times 10 \times \sin 105^\circ \\ &= 33.81 \text{ m}^2 \quad (2 \text{ dec pl}) \end{aligned}$$

$$\text{d) Given point : } (-2, 2)$$

$$\text{Given line : } 6x + 3y - 1 = 0$$

Perp. dist.

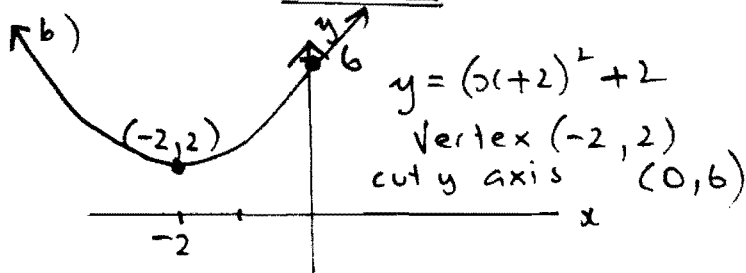
$$\begin{aligned} d &= \frac{|(6 \times -2) + (3 \times 2) - 1|}{\sqrt{6^2 + 3^2}} \\ &= \frac{7\sqrt{5}}{15} \text{ units} \end{aligned}$$

Question 4

a) $\sin 80^\circ = \cos (90-x)$

$80 + (90-x) = 90$

$\therefore x = 80^\circ$



c)

$$43 + 59 + 21 + y = 180^\circ$$

(cointerior angles $AB \parallel CD$)

$$\therefore y = 57^\circ$$

$$59 + x + 57 = 180^\circ$$

(angle sum of $\triangle AEC$)

$$x = 64^\circ$$

Question 5

a) $2x+1 = 3x-2$ & $2x+1 = -(3x-2)$

$$3 = x \qquad 2x+1 = -3x+2$$

$$5x = 1$$

$$x = 1/5$$

check: $|6+1| = 9-2$, $|1\frac{2}{5}| \neq -1\frac{2}{5}$

$$7 = 7$$

true false

$\therefore x = 3$ only solution

b)

$$\frac{AC}{\sin 45^\circ} = \frac{5}{\sin 60^\circ}$$

$$AC = \frac{5 \sin 45^\circ}{\sin 60^\circ}$$

$$= \left(5 \cdot \frac{1}{\sqrt{2}}\right) \div \frac{\sqrt{3}}{2}$$

$$= \frac{10}{\sqrt{6}} \text{ cm}$$

c)

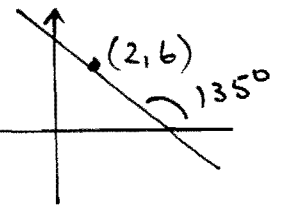
$m = \tan 135^\circ$

$m = -1$

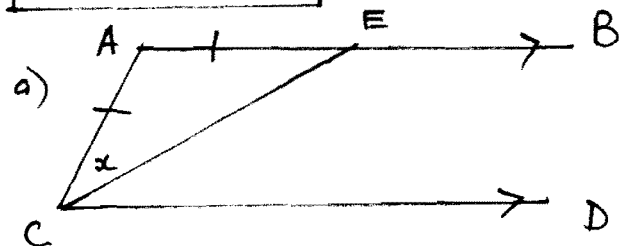
$y - 6 = -1(x - 2)$

$y - 6 = -x + 2$

$x + y - 8 = 0$



Question 6



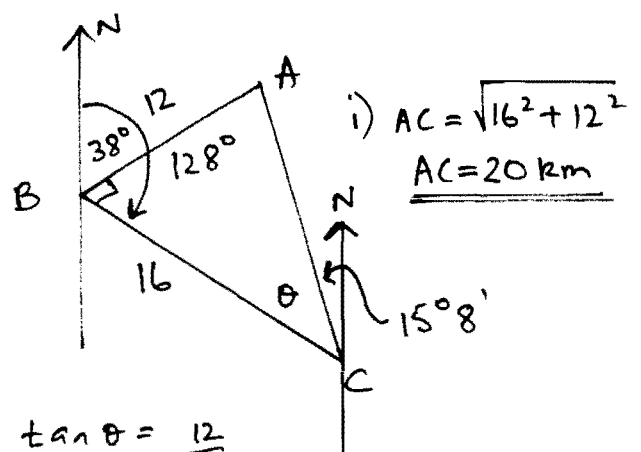
Let $\hat{ACE} = x$

$\therefore \hat{AEC} = x$ (base angles of isosceles triangle)

$\hat{ECD} = x$ (alternate angles)

$AB \parallel CD$

$\therefore \hat{ACE} = \hat{ECD}$



ii) $\tan \theta = \frac{12}{16}$

$\therefore \theta = 36^\circ 52'$

$\therefore \text{Bearing } 344^\circ 52'$

$$\begin{aligned}
 \text{c) } 5x - y - 2 &= 0 \\
 y &= 5x - 2 \quad m = 5 \quad (3, 7) \\
 y - 7 &= 5(x - 3) \\
 y - 7 &= 5x - 15 \\
 \underline{0 &= 5x - y - 8}
 \end{aligned}$$

Question 7

$$\text{a) } \tan 20^\circ = \frac{SO}{AC}$$

$$\text{i) } AC = \frac{SO}{\tan 20^\circ}$$

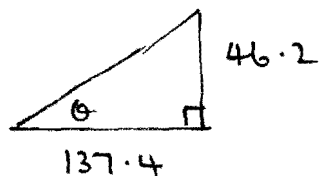
$$AC = 137.4 \text{ m}$$

$$\text{ii) } \tan 35^\circ = \frac{CD}{137.4}$$

$$CD = 137.4 \cdot \tan 35^\circ$$

$$\underline{CD = 96.2 \text{ m}}$$

iii)



$$\tan \theta = \frac{46.2}{137.4}$$

$$\underline{\theta = 18^\circ 35'}$$

$$\text{b) } M(1, 1) \quad m = -2$$

$$y - 1 = -2(x - 1)$$

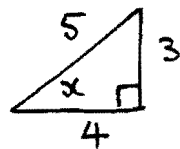
$$y - 1 = -2x + 2$$

$$\underline{2x + y - 3 = 0}$$

$$\begin{aligned}
 \text{c) } \cos 26^\circ &= \cos (180^\circ + 45^\circ) \\
 &= -\cos 45^\circ \\
 &= -\frac{1}{\sqrt{2}}
 \end{aligned}$$

Question 8

$$\text{a) } \begin{array}{c|c} S & A \checkmark \\ \hline \checkmark T & C \checkmark \end{array}$$



$$\text{i) } \tan x = -\frac{3}{4}$$

$$\text{ii) } \underline{\sec x = \frac{5}{4}}$$

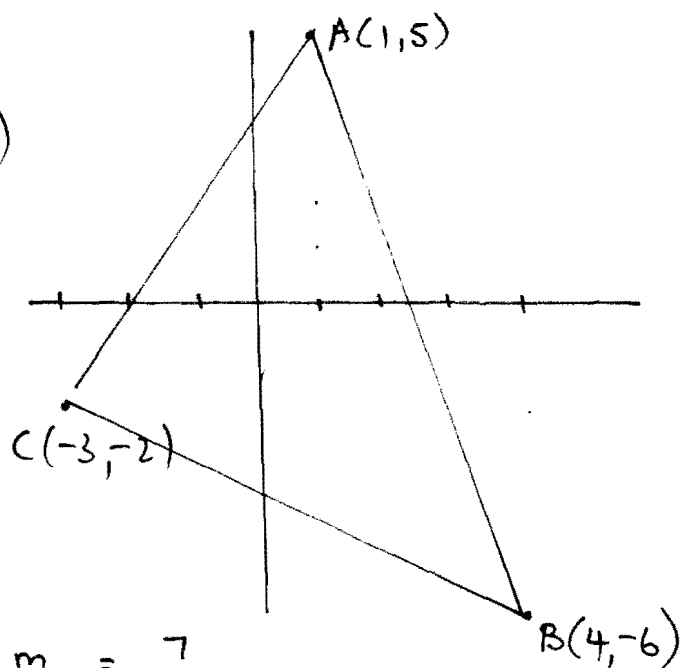
$$\text{b) } \sqrt{3} \tan \theta = 1$$

$$\tan \theta = \frac{1}{\sqrt{3}} \quad \checkmark \begin{array}{c|c} S & A \checkmark \\ \hline \checkmark T & C \end{array}$$

$$\text{acute } \theta = 30^\circ$$

$$\therefore \underline{\theta = 30^\circ, 210^\circ}$$

c)



$$m_{AC} = \frac{7}{4}$$

$$m_{CB} = -\frac{4}{7}$$

$$\text{Since } m_{AC} \cdot m_{CB} = -1$$

$$\therefore AC \perp CB$$

$$\hat{ACB} = 90^\circ \quad \Delta ABC \text{ is } 90^\circ \Delta$$